

RESEARCH ARTICLE

Drastic Circuit Depth Reductions with Preserved Adversarial Robustness by Approximate Encoding for Quantum Machine Learning

Maxwell T. West^{1*}, Azar C. Nakhl¹, Jamie Heredge¹, Floyd M. Creevey¹, Lloyd C. L. Hollenberg¹, Martin Sevier¹, and Muhammad Usman^{1,2*}

¹School of Physics, The University of Melbourne, Parkville, 3010 VIC, Australia. ²Data61, CSIRO, Clayton, 3168 VIC, Australia.

*Address correspondence to: westm2@student.unimelb.edu.au (M.T.W.); musman@unimelb.edu.au (M.U.)

Quantum machine learning (QML) is emerging as an application of quantum computing with the potential to deliver quantum advantage, but its realization for practical applications remains impeded by challenges. Among these, a key barrier is the computationally expensive task of encoding classical data into a quantum state, which could erase any prospective speedups over classical algorithms. In this study, we implement methods for the efficient preparation of quantum states representing encoded image data using variational, genetic, and matrix product state algorithms. Our results show that these methods can approximately prepare states to a level suitable for QML using circuits 2 orders of magnitude shallower than a standard state preparation implementation, thus drastically reducing circuit depth and gate count without unduly sacrificing classification accuracy. Additionally, the QML models trained and evaluated on approximately encoded data displayed an increased robustness to adversarially generated input data perturbations. This partial alleviation of adversarial vulnerability, made possible by the retention of the meaningful large-scale features of the data despite the “drowning out” of adversarial perturbations, constitutes a considerable benefit for approximate state preparation in addition to lessening the requirements of the quantum hardware. Our results, based on simulations and experiments on IBM quantum devices, highlight a promising pathway for the future implementation of accurate and robust QML models on complex datasets relevant for practical applications, bringing the possibility of NISQ-era QML advantage closer to reality.

Introduction

The incredible capabilities of transformer-based models [1–5] have provoked society-wide interest in artificial intelligence (AI) and machine learning (ML), which are increasingly moving beyond academic and scientific applications and into business, industrial, and military use cases. Concurrently, the emergence of programmable quantum computers has led to intense interest in the prospect of quantum machine learning (QML) [6–17]—the study of ML algorithms that exploit the capabilities of quantum computers. Given the rapid proliferation of ML technology, any speedups, enhancements to robustness, or other advantages that can be afforded by quantum computing have the potential to be highly impactful. Indeed, QML models can in principle make use of classically intractable features of data to outperform conventional classical methods through exponential speedups [15] and enhanced resilience to adversarial attacks [18]. However, it is unclear whether such features will generally prove useful for generic classification tasks, particularly classical data that have no inherently quantum mechanical source or structure. Before it can be processed by a quantum computer, such data must first be encoded into a quantum state, a generically $\mathcal{O}(2^{n_{\text{qubits}}})$ process [19] that has

the potential to dominate the runtime of a QML algorithm and negate any potential quantum advantage (Fig. 1A). Quantum state preparation [19–23] is therefore often the first and most computationally expensive subroutine of a QML model but remains a comparatively understudied component of the QML pipeline. With the quantum devices of the current generation offering only limited capabilities in terms of both number of qubits and gate fidelities, the implementation and benchmarking of efficient quantum state preparation techniques (reducing circuit depths) such as performed here is an important step toward improving the prospects of QML models for datasets of practical interest.

Methods

Matrix product state encoding

Matrix product states (MPSs) are a tensor network representation of quantum systems that have found wide application in the simulation of quantum algorithms as a result of their ability to represent and manipulate states of relatively low bipartite entanglement in a memory efficient manner [24]. States represented in MPS form may be decomposed in such a way that they are disentangled entirely using sequential k -local operators

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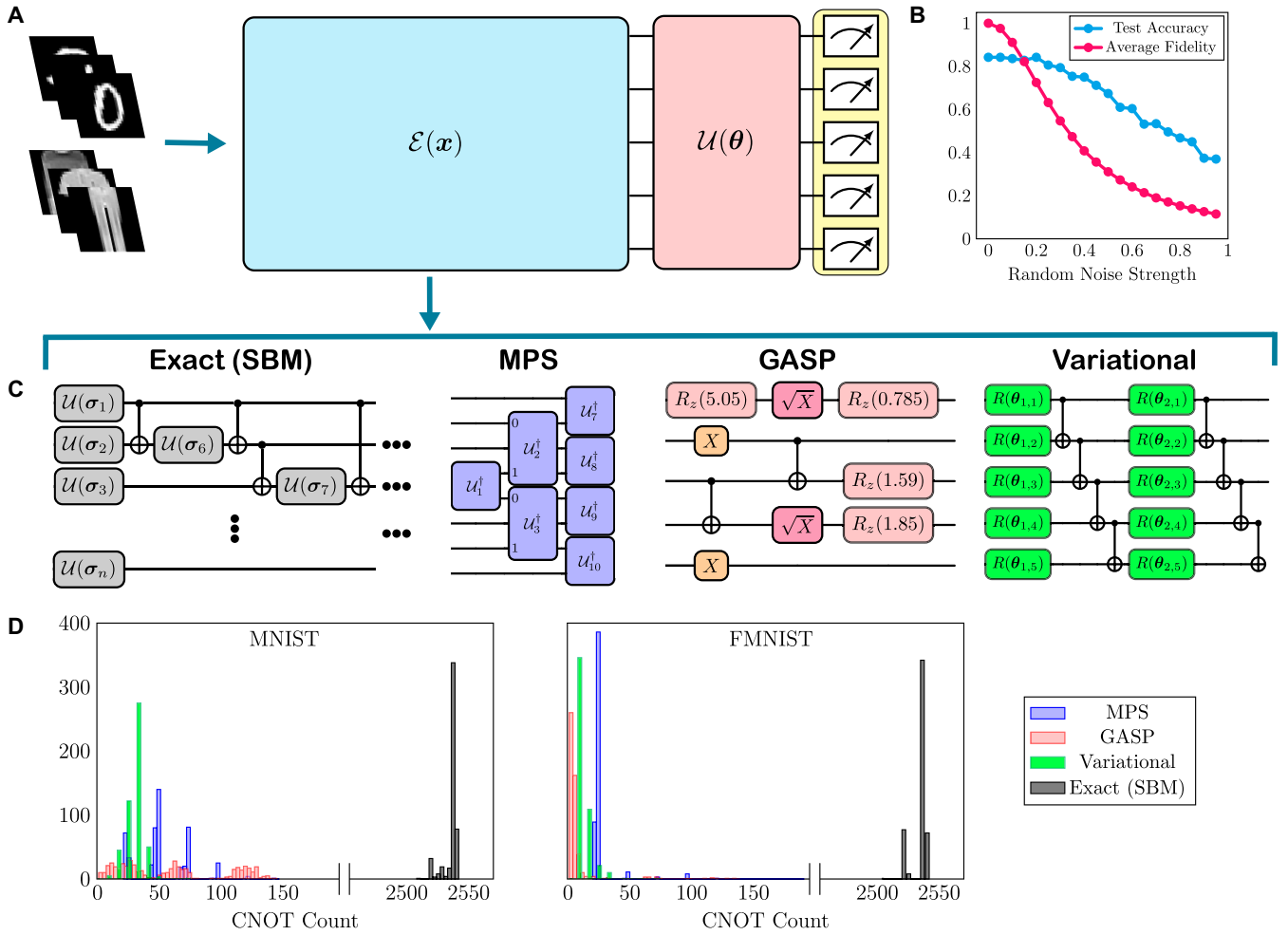


Fig. 1. Approximate state preparation techniques. (A) Schematic diagram of a typical quantum machine learning (QML) framework. Generic n -qubit quantum state preparation $[\mathcal{E}(x)]$ has an $\mathcal{O}(2^n)$ runtime, and the depths of variational circuits $[\mathcal{U}(\theta)]$ are commonly restricted to $\mathcal{O}(\text{polylog } n)$ to avoid barren plateaus [81–84]. Without mitigation, therefore, for large n , the initial state preparation stage dominates the runtime of the algorithm and eliminates any potential quantum advantage afforded by the variational model. (B) The accuracy of a quantum variational classifier on 500 test examples from the MNIST dataset is plotted in the presence of uniformly random perturbations to the image pixels, as is the average fidelity of the resulting encoded states (see Eq. 4) with respect to the clean images. We observe considerable robustness to these perturbations, with the accuracy remaining roughly constant even as the fidelity of the noisy states drops to 60%. (C) Typical circuits resulting from the Shende–Bullock–Markov (SBM) technique employed in the Qiskit state preparation routine as well as the matrix product state (MPS), genetic algorithm for state preparation (GASP), and variational methods. Whereas the SBM, MPS, and variational approaches produce circuits with a deterministic structure, GASP can find very different looking circuits for different target states, occasionally containing only a few CNOT gates. (D) Histogram distributions of the number of CNOT gates required to construct 500 9-qubit states from the MNIST and fashion MNIST (FMNIST) datasets using the methods considered here. The approximate methods were given a target fidelity of 60% and required 2 orders of magnitude fewer gates than the method that Qiskit [19,32] employed to construct the state exactly.

[25,26], where $k = \lceil \log_2(D) \rceil + 1$ is a function of the maximum internal bond dimension of the state D . This procedure has applications in quantum state tomography [27] and image classification [28], and when run in reverse reconstructs the original state using only k -local gates.

The procedure begins by determining the eigen-decomposition of the reduced density matrix over the first k sites,

$$\rho_{(0,1,\dots,k)} = \sum_{i=1}^{2^k} \lambda_i |\Phi_i\rangle\langle\Phi_i|. \quad (1)$$

Reduced density matrices can be determined efficiently given an MPS in canonical form [29]. Moreover, the rank of the reduced density matrix is bounded by 2^{k-1} [27], and consequently, $\lambda_i = 0$ for all $i > 2^{k-1}$. Given this, following Eq. 1, one can construct the following unitary operator,

$$U_0 = \sum_{i=1}^{2^k} |i\rangle\langle\Phi_i|, \quad (2)$$

where the eigenvectors $|\Phi_i\rangle$ of $\rho_{(0,1,\dots,k)}$ are ordered in decreasing order of eigenvalue. Acting with U_0 on the system gives

$$U_0 \rho_{(0,1,\dots,k)} U_0^\dagger = \sum_{i=1}^{2^{k-1}} \lambda_i |i\rangle\langle i|. \quad (3)$$

Hence, we found that the operator U_0 completely disentangles the first qubit from the rest of the system. This procedure was then repeated for subsequent qubits until the entirety of the system became a product state.

This procedure successfully prepares a set of unitary operators that create the desired state; however, the size of the unitary

operators is generally exponentially large, as dictated by the maximum bond dimension D of the MPS. If we restrict ourselves to only considering the case $k = 2$ irrespective of the entanglement of the state, then in general the reduced density matrices will be of full rank and hence Eq. 3 will no longer hold as there will be 2^k nonzero eigenvalues. However, it can be demonstrated that the application of the operator in Eq. 2 in general results in an increase in the magnitude of the amplitude of the $|0\rangle^{\otimes N}$ state [30]. Given this, one may define a heuristic procedure where Eqs. 1 to 3 are performed across the system multiple times with a fixed k until some threshold fidelity is met.

Another consequence of this approximate disentangling procedure is that one can relax the requirement that each subsystem is processed in sequence and instead several subsystems may be considered simultaneously, with the only requirement being that there must be an overlap between all the subsystems by the end of the iteration. As such, we propose a circuit structure where all pairs of adjacent qubits are processed in parallel, with subsequent operations that join each of the disparate subsets (Fig. 1C). We used $k = 2$ as the size of our unitary operators for the entirety of this study as they may be decomposed with at most 3 CNOT gates [31]. The individual 2-qubit unitary operators were decomposed using IBM's Qiskit SDK [32].

Genetic algorithm for state preparation

Genetic algorithms are a classical optimization technique that aims to mimic the process of natural selection in nature [33] and have been used with varying degrees of success in quantum state preparation on quantum computers [23,34,35]. The main advantage of genetic algorithms for state preparation is their ability to reduce the number of CNOT gates required in a state preparation circuit, as well as reduce the total gate count. This results in circuits that are more noise resistant, an important feature in the current NISQ era of quantum computing. The method used in this paper is the genetic algorithm for state preparation (GASP), introduced in [23].

The method begins by defining a target state vector $|\psi_{\text{target}}\rangle$ to be evolved, in this case a state representing an encoded image as in Eq. 4. For this given target state vector, a population of individuals, $|\psi(\theta)_p\rangle$, is produced, with a certain number of "genes", which represent quantum gates, drawn from a specified gate set, in this case, $\{X, \sqrt{X}, R_x(\theta), R_y(\theta), R_z(\theta), \text{CNOT}\}$ where CNOT gates can only be placed between nearest neighbor qubits. The fitness of each individual in this population is then assessed with a defined fitness function, in this case, $\langle\psi_{\text{target}}|\psi(\theta)_p\rangle|^2$. The genetic operators "crossover" and "mutation" are then applied to the population. Crossover in this case is a 1-point crossover, taking half the genetic information (quantum gates) from one parent, and half from the other. Mutation corresponds to changing each gene (gate) in an individual with probability $p = 5\%$. Sequential least squares programming optimization [36] is then run on each individual to allow the highest fidelity for each individual (circuit structure) to be achieved. The genetic operator "selection" is then applied to the population, which in this case is roulette wheel selection, in which each individual is assigned a probability of selection based on its fidelity, with the fittest individuals having the highest probability of selection. This method allows individuals that are low-fitness but "lucky" to be selected, thus maintaining a high degree of genetic diversity. This iteratively increases the fitness of the population. These steps are repeated until the

target fitness is achieved, or there have been 200 generations without the desired fitness achieved. If the target fitness is achieved, the number of gates in the circuit is reduced, so as to achieve the shortest circuit required to produce the target state vector. If the target fitness is not achieved, the number of genes in the individuals is increased, allowing more complex states to be evolved. This increase and decrease of genes is implemented in a binary search (starting from an initial population with $N_{\text{genes}} = 64$) to exponentially reduce the complexity of finding the optimal number of genes compared to a linear search starting from $N_{\text{genes}} = 1$.

Unlike the other methods considered in this study, the circuits discovered by the GASP algorithm were not restricted to be of a repeating, deterministic form (Fig. 1C). This allowed the algorithm to occasionally find circuits consisting of only a few CNOT gates, even for complicated 9-qubit states (Fig. 1D), by exploiting the specific details of individual state vectors.

Variational encoding

Our final method consists of a variational circuit $\mathcal{V}(\theta)$, which was optimized to maximize the fidelity $|\langle\psi_{\text{target}}|\mathcal{V}(\theta)|0\rangle|^2$ of the resulting state with the target state $|\psi_{\text{target}}\rangle$. The variational circuit is composed of a sequence of layers, with each layer consisting of an arbitrary rotation on each qubit i , followed by a chain of CNOT gates applied between adjacent qubits (Fig. 1F). Initially, $\mathcal{V}(\theta)$ is composed of a single such layer, and its parameters are optimized with 100 steps of the Adam optimizer [37]. If this fails to produce a state that reaches the given target fidelity, then an additional layer is added and the optimization is repeated (up to a maximum of 25 layers). Despite the simplicity of this method, we found that it was able to prepare the target states using resources commensurate with our other 2 approaches, generally requiring only a few variational layers to achieve 60% fidelity (Fig. 1C).

Results and Discussion

This study demonstrates that one can preserve the accuracy and increase the adversarial robustness of QML models classifying image data while simultaneously reducing the encoding circuit complexities by orders of magnitude by moving to approximate data encoding schemes, which thus provide a crucial advantage in experimentally implementing QML on noisy quantum hardware platforms. The ability to do this stems from the generic resilience of ML models to random perturbations of their inputs [38]. For example, Fig. 1B shows that our QML models are capable of maintaining their accuracy on noisy image data (with fidelities as low as 60%) encoded exactly into quantum states. This indicates that an approximation to the input quantum state compromising fidelity but resulting in easier circuit preparation with reduced depths (speeding up the state preparation) may have only a minor impact on QML accuracy. Motivated by this remarkable property, we considered 3 independent approximate state preparation methods (Fig. 1C), based on MPSs [25–27], the GASP of [23], and variational circuits. Our results show that all 3 methods are capable of preparing states representing images from the standard modified National Institute of Standards and Technology (MNIST) [39] and fashion MNIST (FMNIST) [40] datasets using circuits 2 orders of magnitude shallower than the high-fidelity states constructed by the exact method, the Shende–Bullock–Markov (SBM) algorithm employed by Qiskit [19,32], as shown in Fig. 1D,

while largely maintaining the accuracies achieved by the exact method. Furthermore, on a simple dataset of 8×8 images, the savings in circuit depth afforded by the approximations were sufficient to perform classification on the device `ibm_qalgiers` with up to 96% accuracy, whereas the exact encoding method led to circuits dominated by noise and performance indistinguishable from random guessing.

The adversarial vulnerability of QML models has recently become a topic of considerable research interest due to the serious concerns resulting from the increasing incorporation of ML in security-sensitive applications [41]. It has been suggested empirically [18] that QML models may provide enhanced resilience to (classical) transferred adversarial attacks and thus offer an advantage to early adopters of quantum computers. Appealingly, this argument is independent of whether the classically intractable features learnt by quantum networks are superior for the classification task at hand, a condition that is by no means guaranteed [18]; thus, there is a strong motivation for studying QML. We found that the adversarial robustness of the approximate models considered in this study is slightly

greater than that of their exact counterparts because errors in the state preparation process “drown out” the carefully constructed adversarial perturbations (Fig. 2A). This situation is reminiscent of well-known studies in classical ML [42–44] in which random noise was intentionally introduced to combat adversarial perturbations. Here, although our primary motivation was to reduce circuit depths, we obtained a similar benefit through the pseudorandomness introduced by approximate state preparation. These results, in which drastic improvements to the circuit depth of the state preparation component of the models led to a slight increase in resilience to adversarial attacks, while almost preserving classification accuracy, highlight a promising pathway for the future development and deployment of QML models in practical applications where security and robustness are key parameters of interest.

Approximate state preparation

Our investigation focused on the problem of image classification, with images drawn from the MNIST [39] and FMNIST [40] datasets (as well as, later, a simple dataset constructed for

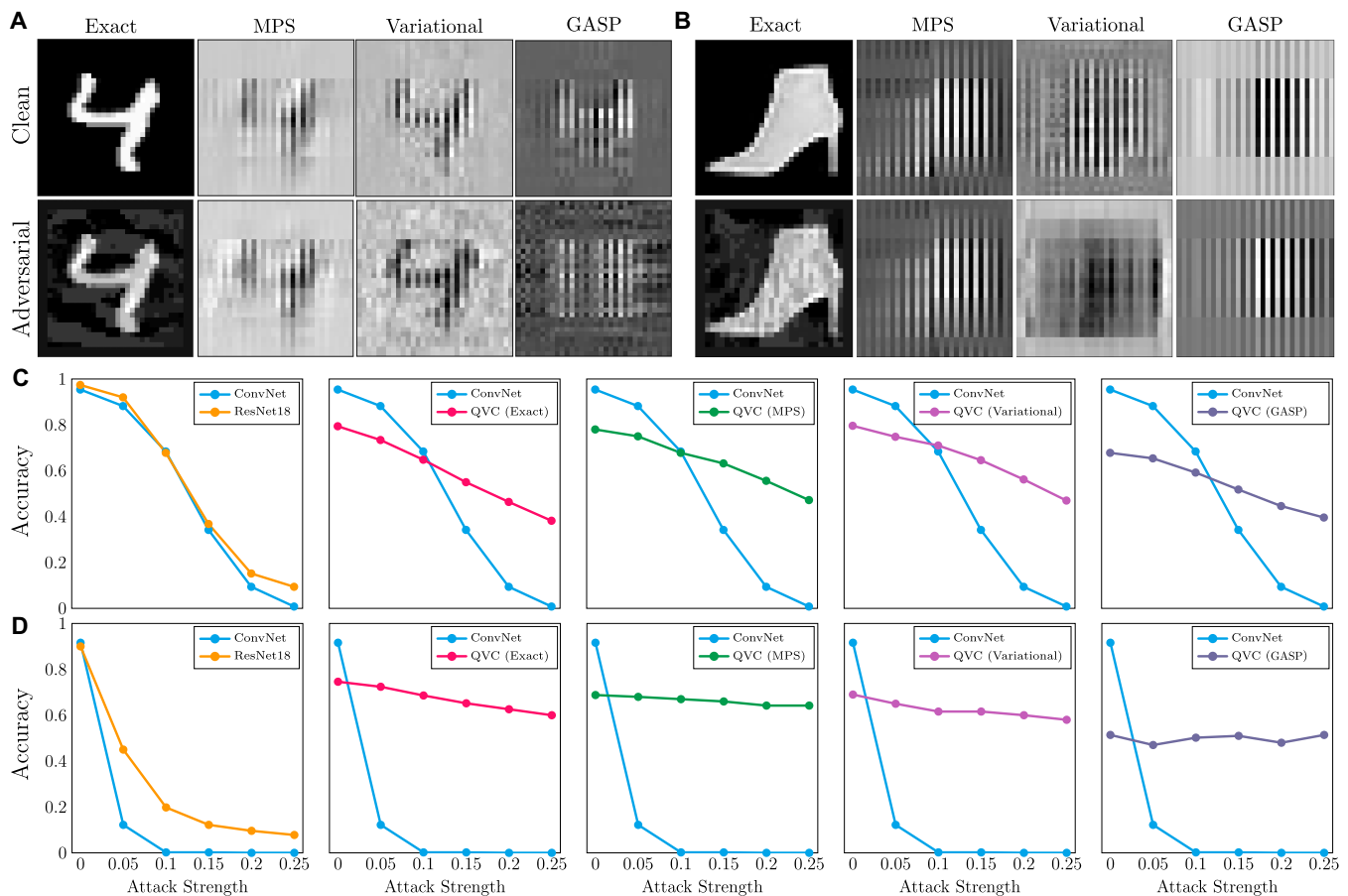


Fig. 2. Adversarially robust QML. (A) Given an approximately prepared state, we can use Eq. 4 to reconstruct the image it represents. Here, this was done for an example image from the MNIST dataset (top left) and a corresponding adversarial example (bottom left). Despite the clear loss of visual quality in the approximate images, the QVCs retain their level of performance to a remarkable degree. This is especially true in the adversarial case; adversarial perturbations are carefully constructed, and a low-fidelity approximation can overlook the fine details that are required to trick the network [43]. (B) The case of the FMNIST dataset is similar. Further examples are shown in Figs. S1 and S2, where slight variations on the encoding scheme are discussed. (C) The accuracies obtained by a classical convolutional neural network (CNN), ResNet18 [71], and the QVCs on adversarial images from the MNIST dataset generated by attacking the CNN. As reported in [18], we found that the QVCs display considerable robustness to the classically generated adversarial images. At an attack strength of zero (i.e., unperturbed images), we found that approximate state preparation techniques can be used to obtain accuracy comparable to that of a model using the exact states. The MPS and variational approaches more readily obtained higher fidelities than GASP, accounting for their improved performance (Figs. S3 and S4). (D) Similarly, we trained the models on the FMNIST dataset and constructed adversarial examples by attacking the CNN. We again found considerably improved robustness on the part of the QVCs.

the purpose of running small QML models on real devices). We will review the details of QML as needed; for a fuller exposition, we refer the interested reader to some standard references [6, 45]. The first step of our QML models is an encoding unitary $\mathcal{E}(\mathbf{x})$, which, given an image $\mathbf{x}$ (represented as a vector of pixel values), creates a quantum state $|\psi(\mathbf{x})\rangle = \mathcal{E}(\mathbf{x})|0\rangle^{\otimes n}$. Due to the high dimensionality of the data (images from MNIST and FMNIST contain 28×28 pixels and so $\mathbf{x} \in \mathbb{R}^{784}$), the current limitations of quantum hardware, and the difficulty of simulating quantum computers possessing large numbers of qubits, we employed the method of amplitude encoding, which is highly qubit efficient compared to other common encoding techniques such as angle encoding [46]. Moreover, as the pixel values are real numbers, we can encode 2 of them into each of the complex amplitudes of the state $|\psi(\mathbf{x})\rangle$. Our mapping procedure is therefore

$$x \mapsto |\psi(\mathbf{x})\rangle = \frac{1}{\|\mathbf{x}\|^2} \sum_{j=0}^{2^n-1} (x_{2j} + ix_{2j+1}) |j\rangle \quad (4)$$

and requires $n = \log_2(28 \times 28) - 1 = 9$ qubits. We note that due to the global phase ambiguity and the normalization constraint of a quantum state, this method of encoding cannot distinguish between 2 images that differ only by a constant factor; however, for the image classification tasks considered here, this did not prove to be an important deficiency. Although this method of encoding is very efficient in terms of the required number of qubits, it involves the preparation of highly entangled states, the circuit depth required for which is generically exponential in the number of qubits. For the $n = 9$ states we considered, the resulting circuit depths of $\sim 2^9 = 512$ put the preparation of $|\psi(\mathbf{x})\rangle$ with high fidelity beyond the capability of the noisy quantum hardware available today. Although the techniques that we present here also appear to scale exponentially in the number of qubits, they do so with considerably lower constant factors, which as we shall see can translate into genuine practical advantages, in particular for image processing tasks that require relatively few qubits even for high-resolution images.

Interestingly, the QML models we employed (Fig. 1A) display considerable robustness to random perturbations of the input states $|\psi(\mathbf{x})\rangle$, which suggests that preparing them with high fidelity is unnecessary. Figure 1B shows the result of adding uniformly random (not to be confused with adversarial) noise to a set of test images from the MNIST dataset before encoding and plotting the accuracy of our model on the resulting images, and the average fidelity of the encoded states of the noisy images with respect to the encoded states of the clean images, as a function of the level of random noise applied. We found that the accuracy of the classifier remains steady even as the fidelities of the noisy states drop to $\sim 60\%$ to 70% . This is an extremely encouraging result, as the preparation of a quantum state to this level of fidelity is a drastically easier task than preparation to high fidelity.

Motivated by the notion that accurate QML models can be implemented on very noisy encoded states, we implemented 3 methods of approximate state preparation and benchmarked them by both raw accuracy and adversarial robustness. The methods are as follows: a deterministic technique for constructing an MPS approximation to the target state [27] and 2 ML methods that attempt to learn unitaries that perform the state preparation, relying in one case on a genetic algorithm (GASP) [23] and in the other on variational quantum circuits. The full

details of each of these approaches are discussed in Methods. When applied to the 9 qubit states representing images from the MNIST and FMNIST datasets, these methods led to encoding circuits with 2 orders of magnitude fewer entangling gates (Fig. 1D) when compared to the (exact) SBM algorithm implemented in Qiskit [19,32].

Adversarial QML

Despite the increasingly impressive successes of ML across a large number of domains, the prospect of automating high-impact tasks with a low tolerance for failure has been hindered by the lack of explainability of ML models, which typically produce their outputs via computational processes that are inscrutable to humans [47]. Among other difficulties, the complicated and opaque nature of the decision-making process of these models leaves them open to surprising failure modes [38]. Indeed, an unexpected discovery of recent years has been the extreme susceptibility of artificial neural networks to careful tampering with their inputs (“adversarial attacks”), which can cause dramatic failures even in sophisticated high-performance models [38,42,48–56]. Moreover, adversarial attacks tend to transfer between neural networks—an adversarial example constructed to deceive a specific network can also fool other, entirely independent networks [38,57–59]. This considerably increases the threat posed by adversarial attacks in practice, as it means that one can trick an external model without being privy to the precise details of its architecture. As part of the ongoing effort to determine the potential benefits and practicality of QML, the adversarial robustness of quantum classifiers has therefore become the subject of much recent attention [18,41,60–67]. The evidence compiled to date indicates that QML models also suffer from the existence of adversarial examples, as a result of counterintuitive geometrical properties of the Hilbert spaces in which their classifications takes place [61]. A recent study [18], however, has suggested that quantum classifiers may demonstrate considerable resilience to adversarial examples that were generated by attacking a classical network and then transferred to the quantum network. This is due to the QML models learning a different set of features to those generically learnt by the classical networks and is potentially a new avenue through which to demonstrate quantum advantage in ML: quantum networks that offer enhanced robustness to classical adversarial attacks [18]. This increased robustness has been demonstrated for both image [18] and radio signal [68] datasets.

Our QML models follow a standard [18] 3-step paradigm: a data-dependent encoding unitary $\mathcal{E}(\mathbf{x})$ (implemented variously by our MPS, variational, and GASP strategies), followed by a trainable variational circuit $U(\theta)$, followed by the measurement of a set of observables $\{M_j\}_{j=1}^{n_{\text{classes}}}$. Having discussed the initial data encoding stage in detail in the previous section, we now turn to the design of the remaining aspects of our models and their susceptibility to adversarially manipulated data.

The variational component of our models consists of a sequence of repeated layers. Each layer comprises (trainable) arbitrary rotations on each qubit, followed by a chain of CNOT gates between neighboring qubits. The prediction $\hat{y}_\theta(\mathbf{x})$ of a model on a data point $\mathbf{x}$ is defined to be the class corresponding to the measured observable with the highest expectation value, i.e.,

$$\hat{y}_\theta(\mathbf{x}) = \arg \max_j \langle 0 |^{\otimes n} \mathcal{E}(\mathbf{x})^\dagger U(\theta)^\dagger M_j U(\theta) \mathcal{E}(\mathbf{x}) | 0 \rangle^{\otimes n}. \quad (5)$$

We can subsequently assign a probability $p(j|\mathbf{x})$ that the input $\mathbf{x}$ belongs to the class with index j via a standard softmax normalization:

$$p_\theta(j|\mathbf{x}) = \frac{\exp(\langle \psi(\mathbf{x}) | U(\theta)^\dagger M_j U(\theta) | \psi(\mathbf{x}) \rangle)}{\sum_k (\exp(\langle \psi(\mathbf{x}) | U(\theta)^\dagger M_k U(\theta) | \psi(\mathbf{x}) \rangle))}, \quad (6)$$

where we have written $|\psi(\mathbf{x})\rangle = \varepsilon(\mathbf{x})|0\rangle^{\otimes n}$ to denote the encoded state corresponding to the data point $\mathbf{x}$. Throughout this study, we take the j th measurement observable M_j to be the Pauli z operator acting on the j th qubit. During training, the variational parameters θ were updated to minimize the empirical cross-entropy loss [69] on the training set,

$$\mathcal{L}_{N_{\text{train}}}(\theta) = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} -\log(p_\theta(y_i|\mathbf{x}_i)), \quad (7)$$

where N_{train} is the number of training examples and y_i is the label of the datapoint $\mathbf{x}_i$. We trained the models by seeking parameters $\theta^* = \text{argmin}_\theta \mathcal{L}_{N_{\text{train}}}(\theta)$ which minimize the empirical loss, where we carried out the optimization using the Adam optimizer [37] with a learning rate of 1×10^{-3} and a training set of 30,000 images. For each state preparation technique, a model was trained and tested on data prepared using that method, with the exception of GASP, which was tested on models trained using variationally prepared states due to limited computational resources. The accuracies obtained on the validation set throughout the training process for each of the considered state preparation methods are shown in Fig. S3.

The process of generating an adversarial example with respect to a given ML model (“an adversarial attack”) consists of an attempt to maximize the loss function [53,58,70]. We considered the model parameters θ to be fixed at their trained values θ^* and, given a target example $\mathbf{x}$ with label y and a region Δ consisting of “small” inputs (with respect to a relevant metric, see below), searched for a perturbation

$$\delta_{\text{adv}} = \arg \min_{\delta \in \Delta} p_{\theta^*}(y|\mathbf{x} + \delta), \quad (8)$$

where y is the label associated with $\mathbf{x}$. That is, we looked for a minor modification we could make to the input $\mathbf{x}$, which would minimize the probability that the model would assign the label y to the new input. If the perturbation is small, then, by assumption, the true label of the modified datapoint is also y , and so if the procedure is successful, the model has been tricked into making a misclassification. Typically, one takes Δ to be the space of inputs with l_p norm $\|\delta\|_p = (\sum_{i=1}^n |\delta_i|^p)^{1/p}$ bounded by a small constant ϵ for some chosen p . In the case of image classification, the l_∞ norm $\|\delta\|_\infty = \max_i |\delta_i|$ is usually selected, corresponding to perturbations whose pixels are each individually bounded in magnitude by ϵ . Such an attack is said to have “strength” ϵ . The l_∞ norm is the metric we considered in this study.

We generated adversarial perturbations using a method called “projected gradient descent” (PGD) [53], which amounts to attempting to solve the optimization problem of Eq. 8 iteratively through a slight modification of standard gradient descent. Specifically, starting uniformly randomly in the space of allowable perturbations Δ , we constructed δ_{adv} via the procedure

$$\delta'_0 \sim \text{Unif}(\Delta), \delta'_{t+1} = \mathcal{P}_\Delta(\delta'_t - \alpha \nabla_{\delta} \log(p_{\theta^*}(y|\mathbf{x} + \delta'_t)))^{(9)}$$

for a given number of iterations and a step size α . After each gradient step, the perturbation was projected back into Δ by the projector $\mathcal{P}_\Delta$. Here, we took 3 steps, i.e., $\delta_{\text{adv}} = \delta'_3$, and set $\alpha = \epsilon/3$, where ϵ is the strength of the attack. We selected PGD due to its “universality” as a first-order attack [53] (i.e., the strongest attack given only access to local first-order information about the network) but also note that, in a previous study [18], different types of adversarial attacks were found to perform quite similarly in this context. It is a particularly notable result of adversarial ML that standard image-classifying convolutional neural networks are vulnerable to being fooled by small perturbations that a human would have no trouble recognizing as not meaningfully changing the image [38] (Fig. 2A and B). A further surprising result is the transferability of adversarial examples—whereas the optimization procedure of Eq. 9 involves maximizing the loss function of a specific network, adversarial examples constructed in this way tend to also fool independent networks, even if their architectures differ. We see this phenomenon in Fig. 2C and D, where adversarial examples constructed with respect to a convolutional neural network also succeeded in deceiving a ResNet [71] model. The quantum variational circuits (QVCs), however, displayed a striking robustness to the classically generated adversarial images, as a result of their learning different representations of the features of the images [18].

The considerable reductions in circuit depth attainable by our approximation techniques without large decreases in classification accuracy are possible as a result of the remarkable robustness of the QML models to (nonadversarial) perturbations (Figs. 1B and 2). Further examples of images corresponding to approximately prepared states from the FMNIST dataset are shown in Fig. S1. Despite the poor visual quality of the images, the models are able to discover and exploit subtle properties of the states that are correlated with the class labels of the corresponding images, enabling them to perform classification with reasonable accuracy (Fig. 2C and D). Ironically, the use of this nonhuman approach to the classification and detection of complicated, high-frequency correlations instead of analyzing the large-scale semantic features of images has been blamed for the emergence of adversarial attacks [57]. The classification-informing features of the noisy images employed by the QVCs are not, however, generally the same as those used by a network trained on the exact images, as a result of the noisy preparation procedure failing to recreate their subtle details. This suggests that QVCs trained and evaluated on noisy images may offer increased robustness to adversarial attacks transferred from a network that uses the exact images, as the noisy QVCs are unable to make use of the features employed by the exact models, which are precisely the features targeted in an adversarial attack. Similar ideas have been proposed in the classical ML literature [42–44]. In our case, this noise-induced resilience combines with the fact that the QVCs naturally learn different classes of features than classical neural networks due to the classical intractability of generic quantum computations [18] to produce models that display considerable robustness to classically generated adversarial attacks (Fig. 2). Additionally, we found that this resilience is present when facing attacks conducted on a QVC, with the approximate models partially resisting the transferred attacks (Fig. S5).

Although MNIST and FMNIST are by the standards of classical ML relatively simple datasets, even with the drastic reductions in state preparation circuit depths afforded by the approximation techniques, it is not possible to implement them on the current generation of quantum processors due to the limitations of the available devices. The methods developed in this study are expected to work for complex images when more capable quantum hardware platforms become available in the next few years. This is also evident from a recent theoretical study [72] where variational quantum circuits were directly applied to scanning tunnelling microscopy images of phosphorous donor impurities in silicon—a dataset with direct relevance to a practical application. In the presence of approximate state preparation methods such as those developed in this study, the main bottleneck is the long variational component that is essential for learning image features and hence dominates the total CNOT count. We can nonetheless explicitly see the progress made toward running such models on a physical quantum computer by conducting noisy simulations with varying degrees of noise, and tracking the performance of the various models. Figure 3A shows the results of noisy simulations of our QML models with a simple noise model that assumes ideal single-qubit gates and 2-qubit gates affected by depolarization noise of variable strength p . Although this model overlooks the

complicated effects present in real systems [73], modeling 2-qubit gates as failing with probability p is a useful first-order approximation. We found that our approximate methods are capable of maintaining their accuracy under noise levels 1 to 2 orders of magnitude greater than an exact model generated using Qiskit's standard state preparation technique.

Although this is an encouraging result, it is also valuable to benchmark our techniques beyond a simulation environment, in the presence of real device conditions. For example, Ren et al. [65] have recently shown that adversarial examples can fool QML models running on real hardware. In order to facilitate testing on the limited quantum devices available today, and as a first step toward demonstrating the potential of enhanced robustness in QML, we introduce a simple 2-class dataset that can be handled by QVCs with only a few variational layers. This dataset, which we refer to as the “Shapes” dataset, consists of 8×8 images of squares and crosses (Fig. 3B). The images in this dataset can be encoded by means of Eq. 4 into the state of a 5-qubit system. We again found that our state preparation methods require considerably fewer CNOT gates than circuits produced by Qiskit's generic transpilation algorithm (Fig. 3C). Notably, these reductions were sufficient to enable us to successfully classify the Shapes dataset on real quantum hardware, in the presence of the associated noise. Results collected on

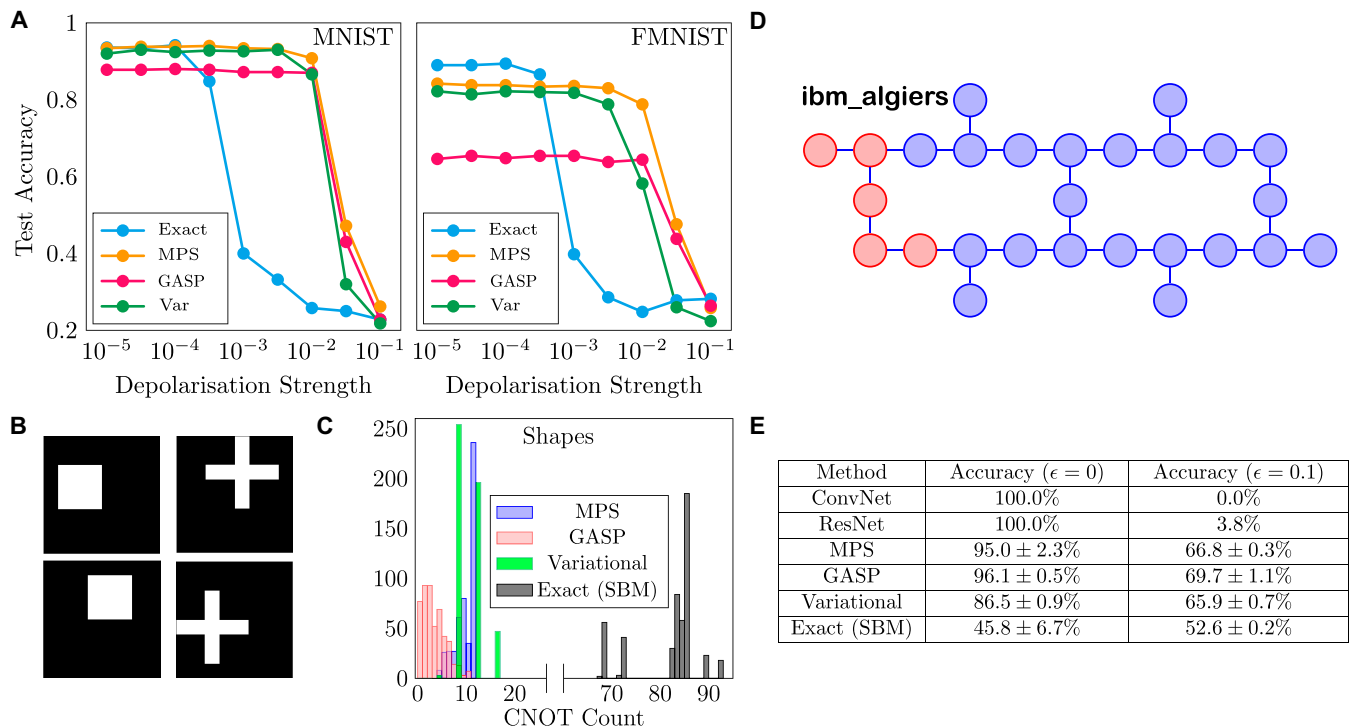


Fig. 3. Experimental demonstration. (A) The accuracies of standard 20-layer QVCs in the presence of depolarization noise acting on each of the 2-qubit gates in the circuit. Due to their shorter depths, the QVCs employing approximate state encoding methods were able to maintain their accuracies up to a more realistic noise level of $p_{\text{depol}} \sim 10^{-2}$. In this example, the QVCs were trained to classify the first 4 classes of the MNIST and FMNIST datasets. (B) We introduced a simple, 2-class dataset “Shapes” consisting of 88 images. Using our encoding scheme (Eq. 4), this allowed us to encode an image into a 5-qubit state. Images from this dataset consist of “squares” and “crosses” positioned randomly on an 88 grid. (C) A histogram of the number of CNOTs needed by the various methods to construct a set of 500 test images from the Shapes dataset to a target of 70% fidelity. As in the cases of MNIST and FMNIST (Fig. 1C), we found that our 3 approximate methods considerably outperformed the default Qiskit behaviour. (D) We evaluated our QML models on this dataset on real hardware, the 27-qubit superconducting system *ibm_algiers*. The topology of *ibm_algiers* is shown, with the 5 qubits employed highlighted in red. (E) Accuracy on 500 test examples from the Shapes dataset for different state preparation methods, evaluated on *ibm_algiers* in the presence of adversarial attacks on ConvNet. We report the mean and standard deviation obtained after running each experiment 3 times. Whereas the default Qiskit state preparation routine produced deep circuits that were overwhelmed by noise, the savings in circuit depth realized by our approximate methods were sufficient to allow our QVC to classify the data with high accuracy. All QVCs again displayed considerable robustness to adversarial attacks generated by attacking a convolutional neural network and transferring the resulting images, but this time on a real device in the presence of complicated hardware noise.

the 27-qubit IBM quantum machine `ibm_algiers` (Fig. 3D) are shown in Fig. 3E, with the approximate methods leading to high classification accuracy and generic state preparation resulting in circuits deep enough to be overwhelmed by noise. The slightly lower accuracy obtained by the variational method may be attributed to the long chains of CNOT gates on neighboring qubits present in the variational ansatz, leading to higher levels of crosstalk noise than experienced in the other methods [74]. On adversarial examples generated by an $\epsilon = 0.1$ attack on ConvNet and transferred to the other listed networks, we observed that the quantum models running on `ibm_algiers` and utilizing approximate state preparation were able to resist the attack more successfully than the classical networks.

Conclusion

The field of QML is rapidly progressing, and multiple proof-of-concept experimental demonstrations have been reported in the literature [65,75–77]. Going forward, the implementation of QML on complex datasets relevant for real-world applications will require steady breakthroughs on several fronts, including the simplification of the complexity of classical data encoding circuits to reduce hardware overheads (such as performed in our study), the design of novel QML architectures capable of learning with superior accuracies, and the implementation of QML models with error mitigation and/or correction protocols. Our study has addressed a key challenge by simplifying arbitrary state preparation, which is an expensive but unavoidable component of many QML algorithms and can be a serious obstacle to the prospect of asymptotic advantages over classical methods. We have argued that the ability of ML models to handle surprisingly large amounts of (nonadversarial) noise may be exploited to considerably reduce the complexity of the required state preparation, by simply settling for low-fidelity approximations to the target states, and letting the model learn to handle the ensuing noise. In this fashion, we have been able to reduce the gate count for complicated 9-qubit states by 2 orders of magnitude compared to exact state preparation while nearly preserving accuracy. Although in some cases the resulting images seem to a human to bear little resemblance to their targets, the QML models were able to learn to classify them correctly. This considerable reduction in gate count drastically reduces the hardware resource requirements for the implementation of fault-tolerant QML based on surface code error correction schemes that typically scale very poorly with the circuit size [41]. Additionally, although all of the methods here are designed to produce increasingly (and arbitrarily) high-fidelity approximations to a given state, by obtaining a low-fidelity approximation simply being a matter of terminating the algorithm early, one could also imagine designing an algorithm that even in principle is incapable of generating greater than (say) 60% fidelity approximations to most states, but in return is able to (typically) find shallower circuits. Such an algorithm would be of great importance to this application.

Furthermore, we have found that the resulting models display an increased resilience to adversarial attacks due to their decreased reliance on subtle, easily exploitable features of the data.

Despite the considerable progress on the state preparation front reported here, there still remain open theoretical questions regarding the scalability of QML models and their resistance to dequantization [78]. For example, the optimal architectures for

QML models remain unclear, and may be problem dependent [14], in contrast to the universal architectures to which classical neural networks have been converging [79,80]. Nonetheless, the ideas explored in this study, combined with the proof-of-concept demonstration on IBM hardware, suggest that there are considerable benefits to sacrificing input state fidelity both for drastically shorter circuit depths on current noisy hardware and for a natural robustness to adversarial tampering. In the immediate future, several commercially available road maps indicate that quantum devices will aggressively scale up, with the possibility of thousands of qubits becoming available within the next 5 to 10 years, which when coupled with the anticipation of error rates that can be overcome by sophisticated error mitigation and correction techniques should allow increasingly capable QML demonstrations. The reduction by orders of magnitude in the encoding circuit resource requirements achieved in our study, which largely retains accuracy and even improves robustness, is an important milestone along the way.

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Author contributions: M.U. conceived the original idea. M.U., M.S., and L.C.L.H. supervised the project. M.T.W. developed the QML framework and contributed to the implementation of approximate encoding schemes. A.C.N., J.H., and F.M.C. implemented MPS encoding, variational encoding, and the GASP method, respectively. All authors contributed to the analysis of the data. M.T.W. and M.U. wrote the manuscript with input from all authors.

Competing interests: The authors declare that they have no competing interests.

Data Availability

The data that support the findings of this study are available within the article. The source code can be provided by the corresponding author upon reasonable request.

Supplementary Materials

Figs. S1 to S5

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Drastic Circuit Depth Reductions with Preserved Adversarial Robustness by Approximate Encoding for Quantum Machine Learning

Maxwell T. Wešt, Azar C. Nakhl, Jamie Heredge, Floyd M. Creevey, Lloyd C. L. Hollenberg, Martin Sevier, and Muhammad Usman

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Quantum machine learning (QML) is emerging as an application of quantum computing with the potential to deliver quantum advantage, but its realization for practical applications remains impeded by challenges. Among these, a key barrier is the computationally expensive task of encoding classical data into a quantum state, which could erase any prospective speedups over classical algorithms. In this study, we implement methods for the efficient preparation of quantum states representing encoded image data using variational, genetic, and matrix product state algorithms. Our results show that these methods can approximately prepare states to a level suitable for QML using circuits 2 orders of magnitude shallower than a standard state preparation implementation, thus drastically reducing circuit depth and gate count without unduly sacrificing classification accuracy. Additionally, the QML models trained and evaluated on approximately encoded data displayed an increased robustness to adversarially generated input data perturbations. This partial alleviation of adversarial vulnerability, made possible by the retention of the meaningful large-scale features of the data despite the “drowning out” of adversarial perturbations, constitutes a considerable benefit for approximate state preparation in addition to lessening the requirements of the quantum hardware. Our results, based on simulations and experiments on IBM quantum devices, highlight a promising pathway for the future implementation of accurate and robust QML models on complex datasets relevant for practical applications, bringing the possibility of NISQ-era QML advantage closer to reality.

Image

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